PATTERN RECOGNITION SEMESTER PROJECT - DEADLINE: 5 NOVEMBER 2026.

Spectral-Spatial Feature Selection for Hyperspectral Land-Cover Classification Using an Enhanced SVM

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Status as of 3 November 2026: all seven methods completed on all five predeclared seeds. Real held-out predictions were independently audited. Results are for a transductive, same-scene random-pixel protocol.

1. Abstract

This study evaluates classical spectral and spatial features for nine-class land-cover recognition on WHU-Hi-LongKou (550 × 400 pixels, 270 bands). Fisher ranking and strict correlation pruning select bands; local means and standard deviations augment them for an RBF SVM (FRSS-SVM). Seven methods share stratified 70/15/15 partitions, with 9,000 capped training pixels and 30,682 held-out test pixels per seed. Validation macro-F1 selects hyperparameters. Across five seeds, FRSS-SVM achieves 97.131 ± 0.201% accuracy and 92.936 ± 0.476% macro-F1. Its macro-F1 difference from the full-spectrum SVM is -0.307 percentage points. MNF-SVM has the highest mean macro-F1 among the evaluated methods. Reported variability is sample SD across splits, not independent-scene uncertainty.

2. Introduction and motivation

Crop mapping can support agricultural inventories and land-use assessment. A hyperspectral pixel contains many closely spaced spectral measurements, which may help distinguish crops but can also be redundant. Spatial context may reduce local ambiguity at the cost of smoothing boundaries. This project examines that trade-off with interpretable feature selection and classical classifiers, avoiding reliance on a GPU.

3. Problem statement

Given a cube $X \in R^{(H \times W \times B)}$, predict one of nine classes for each labelled pixel. The scene's label 0 denotes unlabelled background and is excluded from learning and evaluation. The task is supervised multiclass classification of an agricultural scene, rather than inference on an independently acquired geographic region.

4. Objectives and research questions

1. Does supervised, redundancy-aware band selection preserve classification quality with fewer inputs?

2. Do local means and standard deviations improve per-class recognition relative to spectral-only features?

3. How do selected bands compare with PCA and noise-adjusted components under a shared training budget?

4. What accuracy, feature-size and runtime trade-offs arise between SVM and Random Forest?

These questions are assessed using the held-out results in Sections 13-16; improvements are not assumed.

5. Related work

Hu et al. introduce WHU-Hi as a benchmark for detailed crop recognition using UAV imagery [1]. The official resource page supplies scene and class metadata [2]. Their published scores are not directly comparable with this project's split and are not reused as our findings.

Chang and Lin describe LIBSVM's support-vector classifiers and multiclass implementation [3]. We use the scikit-learn SVC interface to that classical solver. Green et al. develop a noise-aware image transformation [4]; our implementation follows the noise-whitening/generalized-eigenproblem interpretation rather than renaming ordinary PCA. Breiman's Random Forest combines randomized tree predictors [5], providing an alternative nonlinear classifier. The proposed contribution is an evaluated combination of established tools, not invention of Fisher ranking, local statistics or SVM.

6. Dataset and measured distribution

The official LongKou MAT files were downloaded from the publisher's ZIP and checked for size and CRC integrity. Actual variable inspection found a 550×400×270 cube, a 550×400 label map and the counts below, matching the official class table [2]. There are 15,458 background and 204,542 labelled pixels. Thus the dataset exceeds 10 attributes, 1,000 instances and 3 classes.

ID	Class	Observed labelled pixels
1	Corn	34,511
2	Cotton	8,374
3	Sesame	3,031
4	Broad-leaf soybean	63,212
5	Narrow-leaf soybean	4,151
6	Rice	11,854
7	Water	67,056
8	Roads and houses	7,124
9	Mixed weed	5,229

The minimum class is much smaller than the maximum, so macro-F1 and class-wise recall complement overall accuracy. No wavelength calibration vector is assumed by the code; spectral plots label band indices. See data/DATASET_VALIDATION.json and data/DOWNLOAD_PROVENANCE.json for inspection and hashes.

7. Preprocessing and leakage prevention

Pixel splits are stratified 70% training, 15% validation and 15% test, with seeds 7, 19, 42, 73 and 101. The common training pool is capped at 1,000 pixels per class for E1-E6, once per seed. Every fitted mean, standard deviation, Fisher score, correlation, PCA covariance and MNF covariance uses this capped set only. E0 uses original training-pool class proportions to provide the correct majority reference. All actual indices and class counts are persisted at execution time.

Validation labels select configurations; the test set is scored after selection. The winning model is retained without a train+validation refit. Scaling does not use full-scene statistics. Full-scene class spectra and colour stretches are descriptive figures only and do not feed learning. Local statistics use measurements, never labels; random-pixel neighbourhoods can nevertheless share measurements across splits, so this protocol is explicitly transductive and geographically optimistic.

8. Mathematical formulation

For band j, with class size n_c, class mean μ_{cj}, population variance σ^2_{cj} and global training mean μ_j:

$$F_j = [\Sigma_c\, n_c(\mu_{cj} - \mu_j)^2] \,/\, [\Sigma_c\, n_c\, \sigma^2_{cj} + \varepsilon].$$

The implementation uses $\varepsilon = 10^{-12}$ and population within-class variances. Bands are ranked descending, then greedily retained if their absolute training Pearson correlation with every previously retained band is ≤ 0.95. Constant bands are excluded. Strictly pruned sets are never padded with redundant bands.

PCA diagonalizes the covariance of training-standardized inputs and retains the largest-variance directions. The maximum eligible component count is decomposed once; smaller prefixes are reused. Component outputs are rescaled using training statistics before SVM fitting.

For MNF, training-neighbour differences $d_{pq} = (x_p - x_q)/\sqrt{2}$ estimate noise under independent equal-covariance noise and locally smooth signal. Both endpoints belong to the capped training set. Differences are expressed in training-standardized units. With $S_n = \mathrm{cov}(d)$, use $C_n = S_n + \alpha \max(\mathrm{tr}(S_n)/B, \text{machine epsilon})I$, $\alpha = 0.001$. Solve $C_x v = \lambda C_n v$, normalize $v^T C_n v = 1$ and sort λ descending. This is noise whitening followed by PCA in the whitened space; large λ identifies high total-to-noise variance. At least 300 eligible pairs are required and at most 20,000 are sampled with the recorded seed. Too few pairs stop E3 clearly. This estimator can confound signal edges with sensor noise and does not constitute instrument-specific noise calibration [4].

For binary SVM subproblems, the soft-margin objective is $\tfrac{1}{2}||w||^2 + C\Sigma_i \xi_i$ subject to $y_i(w^T\varphi(x_i) + b) \geq 1 - \xi_i$ and $\xi_i \geq 0$. The RBF kernel is $K(x,x') = \exp(-\gamma||x - x'||^2)$; LIBSVM combines binary

subproblems for multiclass prediction [3]. Random Forest predicts through an ensemble of randomized classification trees [5].

9. Proposed FRSS-SVM

For each selected band, compute a 3×3 reflected-border local mean m and population standard deviation s. Concatenate z=[x,m,s], giving 3k features for k bands; 30 bands produce 90 features. Statistics are computed only for selected bands. Variance uses double-precision accumulation with roundoff negatives clamped to zero before square roots. Train-only scaling precedes the same RBF-SVM family used in the principal baseline.

10. Experimental setup and hyperparameters

Method	Feature candidates	Model candidates
E0	Majority prior, no informative features	Dummy most frequent
E1	All bands	RBF SVM
E2	PCA: 20, 50 components	RBF SVM
E3	MNF: 20, 50 components	RBF SVM
E4	15, 30, 45, 60 selected bands	RBF SVM
E5	Selected bands with mean/std: 45, 90, 135, 180 features	RBF SVM
E6	Same spatial feature candidate family	RF: 100, 200 trees

SVM uses C={1,10}, γ={scale,0.01}; RF uses depth≤24 and leaf size≥2. All searches maximize validation macro-F1, with deterministic ties favoring earlier smaller counts. Infeasible dimensions are skipped. If no requested band count survives strict pruning, the entire nonempty pruned set is the predeclared fallback. An optional linear SVM is disabled by default.

The predeclared upper budget is 305 candidate fits. Strict feature feasibility reduced the actual search to 155 fits, all completed. One worker, bounded BLAS threads, a 256-MB SVM cache and 4,096-row prediction batches constrain resources. Four bounded workers predict independent batches; classifier fitting and BLAS remain at one thread. Serial/parallel prediction equality was regression tested. The measured environment is Windows, Python 3.12.5, Intel Core i5-12450H, approximately 16 GB RAM; GPU acceleration was not used. Exact dependencies, source hashes, dataset hashes and configuration are preserved in report/evidence/manifest.json and requirements-windows-lock.txt.

11. Spatial sensitivity protocol

An additional, opt-in block protocol assigns whole 40×40 pixel blocks to partitions with the configured proportions, then erodes each region by a 2-pixel buffer. Neighbourhood footprints are separated and unit checked. Block splits are not guaranteed class stratification or exact pixel proportions. Missing-class partitions raise an error rather than retrying seeds based on performance. A preflight check on the real scene found missing validation classes for seeds 7, 19 and 73; seeds 42 and 101 were feasible. No block classifiers were trained, and seeds were not replaced or selectively reported. Thus no complete five-seed spatial sensitivity claim is possible with this predefined block design. Random-pixel findings must not be described as independent geographic generalization. Details are in report/evidence/protocol_feasibility.json.

12. Evaluation metrics

Overall accuracy is correct predictions divided by test pixels. Class precision is TP/(TP+FP), recall TP/(TP+FN), and F1 their harmonic mean. Macro metrics give equal weight to the fixed nine-class ordering. Undefined ratios become zero consistently. Raw and row-normalized confusion matrices use true rows and predicted columns. Five-run means and sample SD (denominator n−1) are computed from saved

measurements; a single run has undefined SD. Completion flags expose missing seeds.

13. Measured test results

Accuracy and macro-F1 are percentages; ± denotes sample SD over seeds 7, 19, 42, 73 and 101. Every method used the same test split within a seed.

Method	Accuracy (%)	Macro-F1 (%)	Features (mean)	Runs
E0 Majority	32.785 ± 0.000	5.487 ± 0.000	1.0	5/5
E1 Full-spectrum SVM	96.967 ± 0.189	93.243 ± 0.362	270.0	5/5
E2 PCA-SVM	97.200 ± 0.263	94.026 ± 0.444	50.0	5/5
E3 MNF-SVM	98.384 ± 0.270	96.665 ± 0.365	20.0	5/5
E4 Selected-band SVM	89.825 ± 0.290	77.947 ± 0.414	6.0	5/5
E5 FRSS-SVM	97.131 ± 0.201	92.936 ± 0.476	18.0	5/5
E6 Spatial Random Forest	95.353 ± 0.148	89.254 ± 0.188	18.0	5/5

The highest mean macro-F1 is MNF-SVM (96.665%). FRSS-SVM differs from the full-spectrum SVM by +0.164 accuracy points and -0.307 macro-F1 points. These are descriptive comparisons; no significance test or independent geographic validation is claimed.

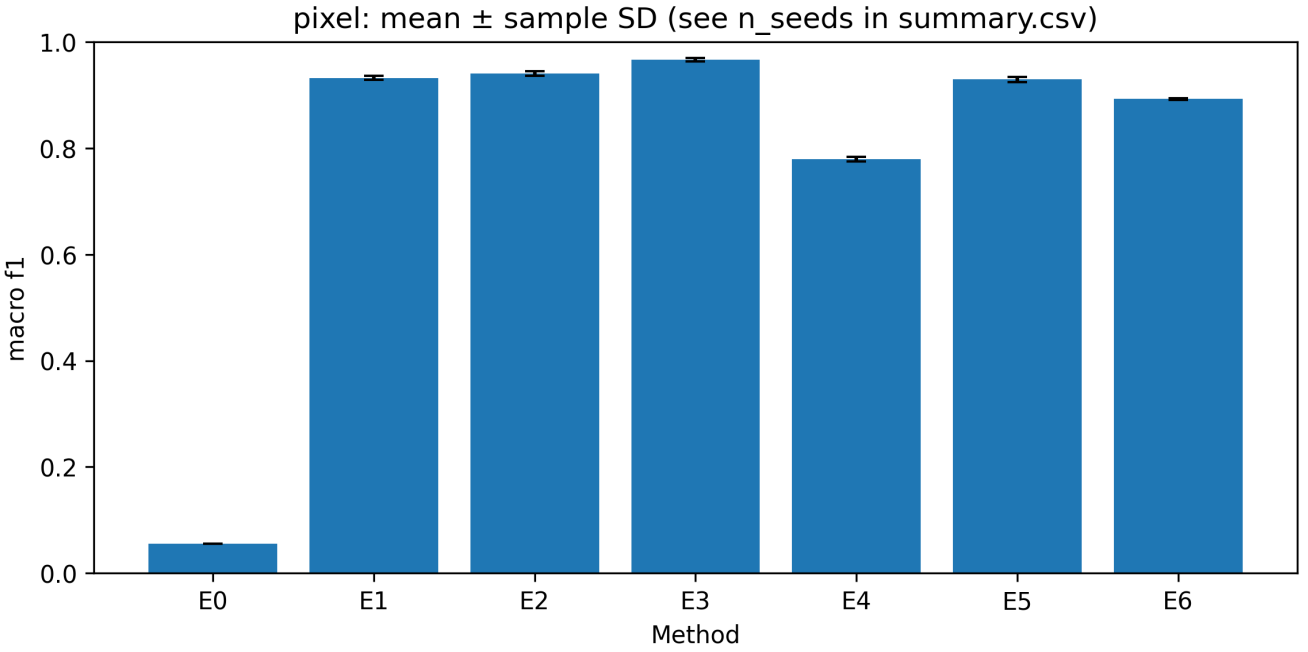


Figure 1. Five-seed test macro-F1 with sample SD.

14. Per-class performance and confusion

The following values are five-seed means in percent. Complete precision, recall, F1 and SD for all seven methods are retained in evidence/per_class_summary.csv.

Class	E1 recall	E5 precision	E5 recall	E5 F1 ± SD
Corn	99.14	99.66	98.13	98.888 ± 0.271
Cotton	96.21	93.66	96.32	94.970 ± 0.393
Sesame	98.33	80.77	98.41	88.699 ± 2.013

Class	E1 recall	E5 precision	E5 recall	E5 F1 ± SD
Broad-leaf soybean	92.38	99.51	93.34	96.324 ± 0.255
Narrow-leaf soybean	96.30	66.57	95.43	78.417 ± 1.578
Rice	99.66	99.65	99.75	99.702 ± 0.103
Water	99.92	100.00	99.95	99.975 ± 0.014
Roads and houses	95.27	94.54	97.31	95.898 ± 0.354
Mixed weed	97.43	73.99	95.97	83.553 ± 1.088

The lowest mean E5 precisions are Narrow-leaf soybean: 66.57% precision versus 95.43% recall; Mixed weed: 73.99% precision versus 95.97% recall. High recall does not imply few false positives. Errors from a large class can noticeably lower a smaller class's precision, which helps explain the gap between overall accuracy and macro-F1.

The largest pooled directed E5 errors are Broad-leaf soybean → Narrow-leaf soybean (1292 predictions); Broad-leaf soybean → Mixed weed (850 predictions); Broad-leaf soybean → Sesame (489 predictions). These counts pool repeated split evaluations, so a scene pixel can contribute more than once. They are not counts of independent pixels across scenes.

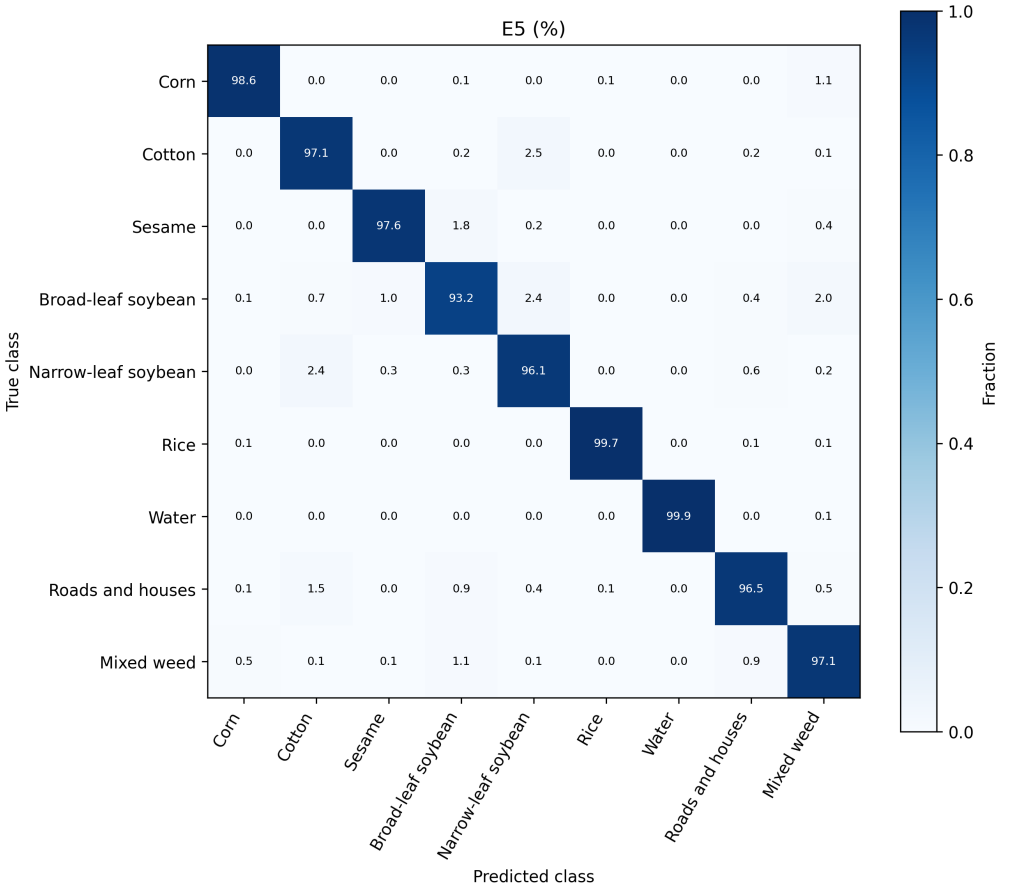


Figure 2. FRSS-SVM held-out confusion matrix for the predeclared display seed 7; annotations are row percentages.

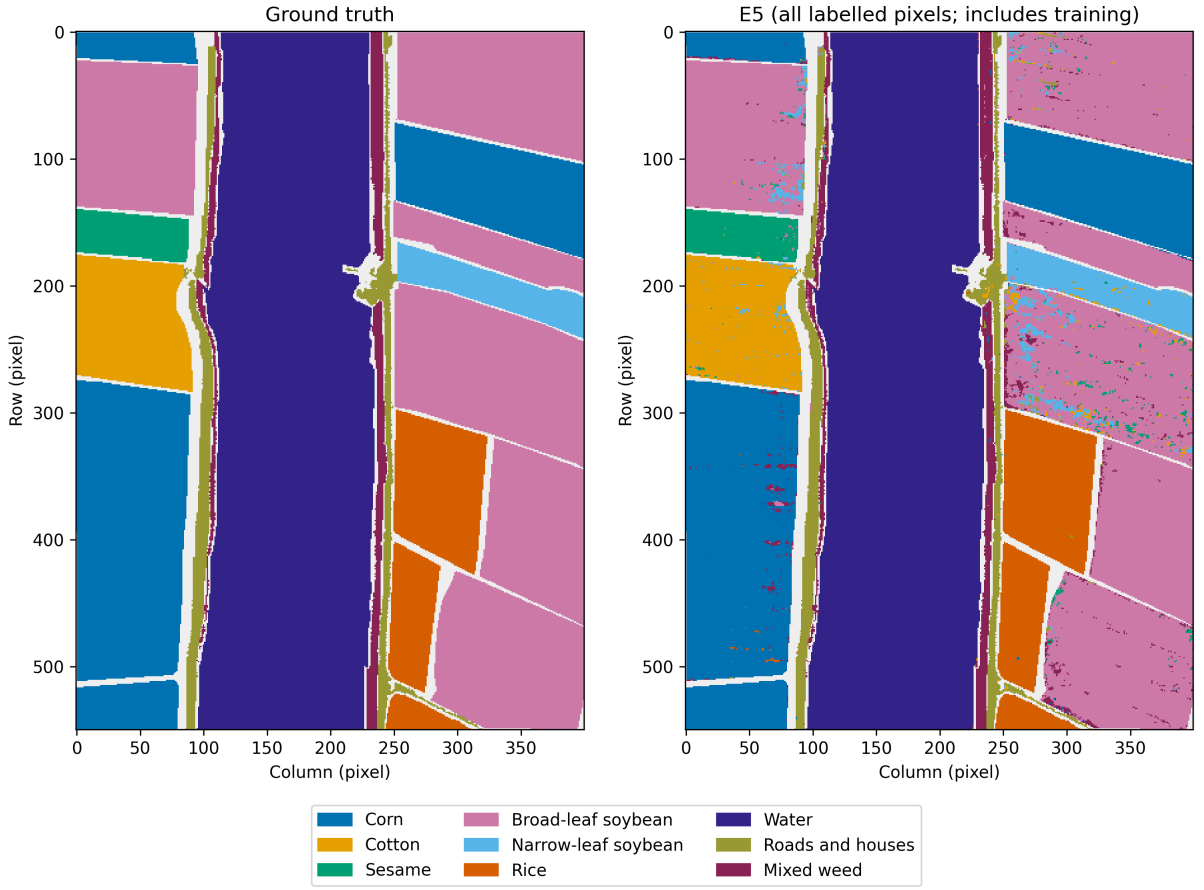


Figure 3. Ground truth and FRSS-SVM descriptive map, seed 7. The full map includes training pixels; only held-out predictions determine reported metrics.

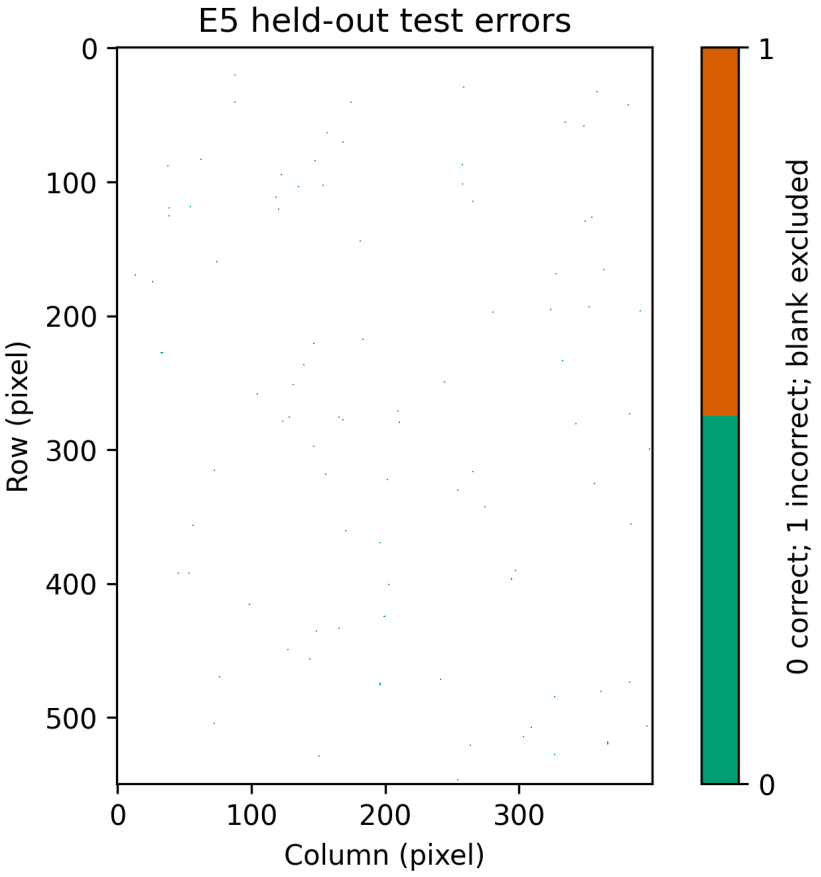


Figure 4. FRSS-SVM errors on held-out test pixels only, seed 7.

15. Paired ablation and selected representations

Differences are proposed minus baseline, in percentage points. Each pair shares the seed and exact split digest.

Comparison	Accuracy difference ± SD	Macro-F1 difference ± SD
E4-E1	-7.142 ± 0.463	-15.297 ± 0.738
E5-E1	0.164 ± 0.330	-0.307 ± 0.749
E5-E4	7.307 ± 0.198	14.990 ± 0.246

Adding local mean/std features to the selected-band family changes macro-F1 by +14.990 ± 0.246 percentage points. Against the full-spectrum SVM the change is -0.307 points, with 93.33% fewer classifier input features on average. Thus recovery of information lost by aggressive band pruning should be distinguished from improvement over all-band classification.

Seed	E2 PCA	E3 MNF	E4 bands	E5 features	E6 features
7	50	20	6	18	18
19	50	20	6	18	18
42	50	20	6	18	18
73	50	20	6	18	18
101	50	20	6	18	18

E5/E6 feature counts include raw bands, means and standard deviations, so their band counts are one third of the displayed count. Strict correlation pruning may leave fewer bands than the smallest nominal request; in that case the predeclared fallback uses all retained bands. This is not a post-test hyperparameter change. Every chosen parameter and zero/one-based band index is recorded in evidence/pixel/. E4/E5 searches tune counts independently, so the ablation compares tuned families rather than an enforced fixed-band intervention.

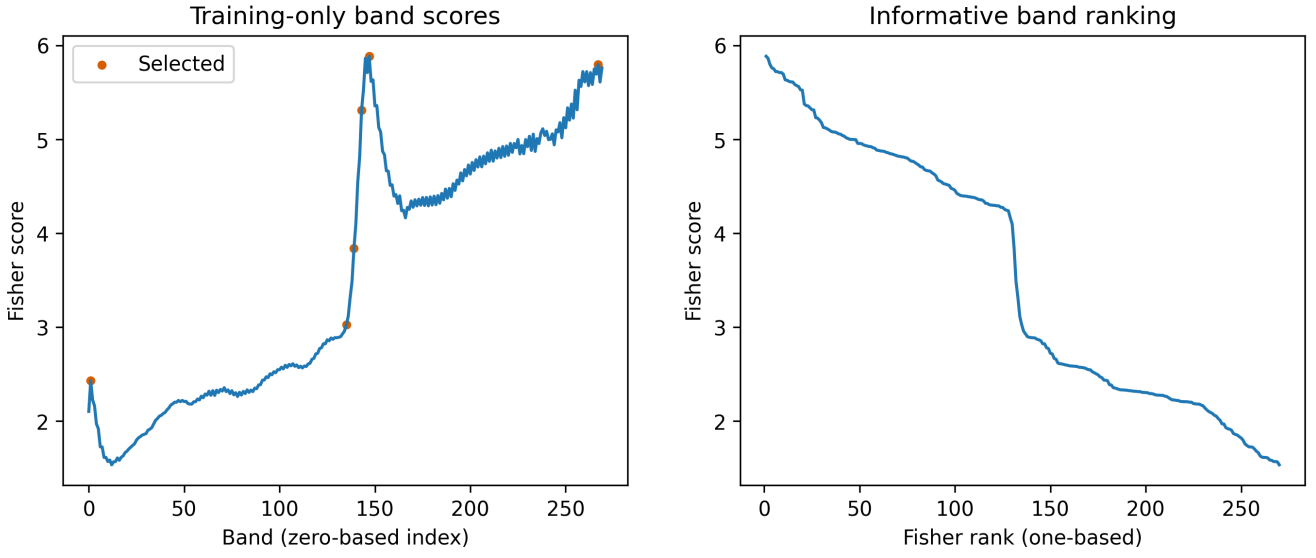


Figure 5. Training-only Fisher scores and selected bands for FRSS-SVM, seed 7.

16. Runtime and dimensionality

Times are seconds, mean ± sample SD. Selected fit is one winning classifier fit; search includes all candidate fits and validation predictions.

Method	Selected fit	Preprocessing	Search	Test prediction
E0	0.028 ± 0.028	0.012 ± 0.010	0.163 ± 0.215	0.110 ± 0.218
E1	5.094 ± 1.461	0.184 ± 0.158	137.535 ± 46.324	17.140 ± 5.372
E2	3.938 ± 1.528	1.134 ± 0.635	122.777 ± 49.886	12.754 ± 6.219
E3	1.505 ± 0.763	0.802 ± 0.366	73.429 ± 25.890	3.620 ± 1.729
E4	1.308 ± 0.373	0.266 ± 0.147	45.328 ± 18.843	5.955 ± 2.645
E5	0.905 ± 0.499	0.267 ± 0.143	28.835 ± 11.275	3.626 ± 1.720
E6	27.876 ± 13.368	0.039 ± 0.067	200.828 ± 374.086	0.903 ± 0.254

Caches reuse decompositions and spatial representations. Preprocessing therefore depends on execution order and is not a cold-start benchmark. Test prediction includes batched transformations but excludes reconstruction of the winning spatial cache. Plotting, disk I/O and full-scene maps are excluded. The computer was also used for documentation and verification, so wall times should be treated as practical observations rather than controlled hardware benchmarks.

17. Limitations and threats to validity

This is a transductive same-scene experiment. Random-pixel neighbours can share measurements across partitions, and geographic autocorrelation can inflate accuracy. Training-only fitting prevents statistical-transform leakage, but does not make spatially neighbouring test pixels geographically independent. The block protocol failed class coverage on three predeclared seeds; no geographic robustness result is claimed. Repeated test sets overlap. Five-run SD captures split sensitivity, not independent-scene uncertainty. The nearly balanced training cap alters class priors. Local noise differences may include signal edges; spatial statistics can blur boundaries. The small validation grid may miss useful settings. The proposed method combines established algorithms and does not introduce a new SVM optimization objective.

18. Conclusion

All 35 method/seed evaluations completed and passed independent prediction-based checks. FRSS-SVM achieved 97.131 ± 0.201% accuracy and 92.936 ± 0.476% macro-F1. Relative to the full-spectrum SVM, its macro-F1 changed by -0.307 percentage points while using 18.0 features on average versus 270. The best observed mean macro-F1 belongs to MNF-SVM. Future work should predeclare a class-covering geographic split, use an independent scene, and compare matched-band spatial ablations.

19. Verified references

1. X. Hu, Y. Zhong, C. Luo and X. Wang. *WHU-Hi: UAV-borne hyperspectral with high spatial resolution (H2) benchmark datasets for hyperspectral image classification*. arXiv:2012.13920, version 2, 2021. Author preprint.

2. Wuhan University RSIDEA. *WHU-Hi resource sharing*. Official dataset and class table, checked 29 October 2026.

3. C.-C. Chang and C.-J. Lin. *LIBSVM: A Library for Support Vector Machines*. ACM Transactions on Intelligent Systems and Technology, 2011. Author-maintained paper, updated 2022.

4. A. A. Green, M. Berman, P. Switzer and M. D. Craig. *A Transformation for Ordering Multispectral Data in Terms of Image Quality with Implications for Noise Removal*. IEEE Transactions on Geoscience and Remote Sensing, 1988. Publisher record, DOI 10.1109/36.3001; indexed original paper scan. Title/authors/year were verified in the indexed original-paper excerpt; the publisher full text and scan fetch were inaccessible in this environment.

5. L. Breiman. *Random Forests*. Machine Learning, 2001. Author-hosted original paper.

These references establish method origins; the performance reported above comes solely from this project's saved experiments. No unverified DOI or accuracy claim is added.

20. Contributions and reproducibility appendix

The team requests an equal (50% / 50%) contribution allocation. The responsibilities below record that allocation; they do not claim separate commit authorship or independently logged execution by each member. Both members share responsibility for reviewing and explaining the complete submission.

Team member	Roll number	Share	Allocated responsibilities
Himanshu Sinha	S20240020290	50%	Problem formulation and related work; Fisher ranking and correlation pruning; spectral/spatial SVM methodology; baseline and ablation interpretation; methodology and discussion sections of the report; joint presentation and final review.
Ghule Yash Mahesh	S20240020288	50%	Dataset provenance and validation; train/validation/test protocol and leakage checks; PCA/MNF and Random Forest comparisons; metric auditing, plots and reproducibility; experimental-results and implementation documentation; joint presentation and final review.

Shared evidence: the implementation modules in src/, the regression suite in tests/, the experiment manifest, per-seed configuration audits, independent audit.json, and the implementation log. These artifacts establish what the project executed; they do not establish individual human authorship.

AI assistance disclosure: Codex assisted with repository review, implementation, automated verification, experiment execution, and drafting the report. Claude Code (Sonnet 5) assisted with resuming and completing the five-seed training run, diagnosing and repairing a concurrent-process artifact from an interrupted run, independent auditing, and generating the final measured report and PDF. Team members should review the code, measured results, interpretation and assigned contribution statements before submission and follow the course's disclosure rules.

The project repository was verified publicly accessible on 2 November 2026: source repository. This local completion has not been pushed; the remote may still contain the earlier version.

From a prepared Python 3.12 environment, run python scripts/download_data.py, then python -m src.main --run --allow-training. In MATLAB, Main starts the complete pipeline and displays summary tables and figures. Exact matching completed runs resume without retraining. A source/configuration mismatch requires a fresh output directory.

Run fingerprint: 58f8e5ab00fe15cd0179328ae11b7572fce15e3f9eee8fd23e1ddad0142f824b. The portable report/evidence bundle retains the manifest, independently verified audit, all per-seed metrics, configuration selections, band indices, class summaries and paired ablations. Large model files and pixel arrays remain under ignored outputs/. The report can be regenerated with python scripts/finalize_report.py, then python scripts/render_report.py.